In PostgreSQL 15 the default value is -1 and a separate parameter vacuum_cost_limit is used instead. The vacuum_cost_limit default value is 200, and this parameter is in effect when VACUUM is run manually from psql. Try running VACUUM VERBOSE trips; from psql. The VERBOSE option is optional but shows more information about what's happening.

Override the default by setting autovacuum_vacuum_cost_limit to a higher value like 2000. With this value, Autovacuum vacuum jobs will complete ten times the work before stopping. This is a good change to try if your Autovacuum Vacuum jobs are not cleaning up dead tuples at a fast enough rate. This might be worth checking mainly for tables with a very high volume of Updates and Deletes. This parameter value can be changed without restarting PostgreSQL so you can adjust it up and down to observe the effect.

Change the value globally in your PostgreSQL config and reload it using pg_ctl reload from your terminal.

sh/postgresql_conf_parameter_change.sh

```
pg_ctl reload -D /path/to/postgresql.conf
server signaled
LOG:  received SIGHUP, reloading configuration files
LOG:  parameter "autovacuum_vacuum_cost_limit" changed to "2000"
```

Although it can be changed globally, a better method to start might be to change it only for tables with heavy Updates and Deletes. To do that, as an example run ALTER TABLE vehicles SET (autovacuum_vacuum_cost_limit = 2000); from psql for the Rideshare database, to adjust the value for the vehicles table.

Another parameter to tune for PostgreSQL versions earlier than 15 is the autovacuum_vacuum_cost_delay. This parameter controls the pause duration. A lower value means Autovacuum is paused for shorter periods of time. In the table below the value is reduced from 20ms to 2ms. In PostgreSQL 15 the default value for this parameter became 2ms, so this parameter change isn't necessary there.

Parameter Name	Original Value	Tuned Value
autovacuum_vacuum_cost_limit	200	2000
autovacuum_vacuum_cost_delay	20ms	2ms

Table 3—Autovacuum Database Parameters

Tuning parameter values

These Autovacuum parameter changes were proposed by PostgreSQL Major Contributor David Rowley[5] after analyzing a PostgreSQL 10 database with

5. https://www.postgresql.org/community/contributors/